Lightweight Streaming Video Models for Real-Time School Violence Detection

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***Abstract*—Early detection of school violence is critical for student safety but remains challenging under real-time surveillance and resource constraints. While recent deep learning approaches achieve high recognition accuracy, most studies overlook deployment feasibility in continuous monitoring scenarios. This work presents a deployment-oriented benchmark of lightweight video models for real-time school violence detection, with a focus on streaming-based architectures. We systematically evaluate representative spatiotemporal models, including MoViNet streaming variants, under a unified training and evaluation protocol. To enhance contextual relevance, we introduce RWF-2000+, an extension of the RWF-2000 dataset incorporating school-oriented surveillance scenarios. Beyond accuracy, we conduct edge-device evaluation on an NVIDIA Jetson AGX Orin, analyzing latency, throughput, and mixed-precision optimization. Results show that lightweight streaming models achieve a favorable accuracy-efficiency trade-off, enabling real-time on-device inference and highlighting the importance of deployment-aware evaluation for practical school surveillance systems.**

***Index Terms*—School violence detection, Real-time video, Lightweight models, Streaming networks, Jetson AGX Orin.**

I. INTRODUCTION

School violence remains a critical societal issue that directly threatens students' physical safety and psychological well-being, with long-term consequences for academic performance, mental health, and social development [1]. Incidents such as physical aggression, bullying, or sudden confrontations often occur without clear warning signs and may escalate rapidly in crowded educational environments. As a result, there is an increasing demand for intelligent surveillance systems capable of early detection and timely intervention to support safer school environments [2].

Conventional security practices based on manual monitoring or rule-based alarm systems are often ineffective in real-world school settings. Human operators are prone to fatigue and subjective judgment, while rule-based approaches struggle to generalize across diverse interaction patterns and environmental conditions [3]. Earlier computer vision techniques relying on handcrafted features, such as optical flow, background subtraction, or trajectory analysis, also exhibit limited robustness under variations in viewpoint, illumination, occlusion, and scene complexity, which restricts their scalability in practical deployments [4].

Recent advances in deep learning have significantly improved automatic violence detection by enabling models to learn spatiotemporal representations directly from raw video data. Architectures such as 3D Convolutional Neural Networks (3D CNNs) and CNN–LSTM hybrids have demonstrated promising performance in capturing motion dynamics associated with violent behavior [5], [6]. However, these models are typically computationally intensive, requiring substantial processing power and memory, which makes them difficult to deploy in real-time school surveillance systems operating under limited hardware and latency constraints.

Despite these advances, many existing studies continue to prioritize recognition accuracy as the primary objective, with comparatively less attention given to computational efficiency and deployment feasibility. Moreover, commonly used benchmarks such as RWF-2000, while effective for evaluating generic violence detection, do not fully capture the temporal characteristics and contextual diversity of authentic school surveillance scenarios. These limitations motivate the need for a more school-oriented benchmark setting and a systematic evaluation that jointly considers recognition performance and runtime efficiency under consistent experimental conditions.

In this work, we investigate lightweight deep learning models for real-time school violence detection, with a particular focus on streaming-based video recognition architectures. Instead of aiming solely for peak accuracy, we emphasize the balance between detection performance and

computational efficiency required for practical deployment. The main contributions of this study are summarized as follows:

- We construct RWF-2000+, an extended dataset that incorporates school-specific surveillance scenarios to improve contextual diversity and real-world relevance.
- We conduct a systematic benchmark of representative spatiotemporal architectures, ranging from conventional 3D CNN and hybrid models to lightweight streaming networks, using a unified training and evaluation protocol.
- We provide empirical evidence that lightweight streaming models, particularly MoViNet-based variants, offer a favorable trade-off between recognition accuracy and computational efficiency, making them suitable for real-time detection of short-duration violent interactions in school surveillance systems.

The remainder of this paper is organized as follows. Section II reviews related work on deep learning–based violence detection. Section III describes the proposed RWF-2000+ dataset and its collection process. Section IV presents the experimental protocol and implementation details. Section V reports quantitative results, runtime analysis, and edge deployment evaluation. Finally, Section VI concludes the paper and discusses potential directions for future work.

II. RELATED WORKS

A. Video Violence Detection Datasets

Datasets serve as the foundational component for video-based violence detection, determining both the generalization capability and stability of deep learning models. Early benchmark datasets, such as the Hockey Fight Dataset, record violent encounters during hockey games. However, their scope and contextual diversity are limited, representing only a narrow form of violence [7]. Other datasets, such as the Movies Fight Dataset, increase diversity by extracting clips from commercial films; nevertheless, their staged nature makes it difficult for models to generalize effectively to real-world surveillance scenarios. The RWF-2000 dataset, introduced by Cheng et al. [8] has become a standard benchmark in this field. It consists of 2,000 real-world surveillance videos labeled as "violent" or "non-violent," significantly improving representativeness and spatiotemporal learning potential . Both Vijeikis et al. [7] and Torre [9] confirmed that RWF-2000 remains one of the most suitable datasets for training and evaluating deep learning models in non-laboratory surveillance conditions. More recently, Kaur and Singh [10] combined the RWF-2000 and RLVS datasets to evaluate a transfer-learning-based ensemble model, demonstrating strong generalization across diverse surveillance contexts. Similarly, Khan et al. [11] conducted cross-domain evaluations involving the RWF-2000, Surveillance Fight, and Industrial Surveillance datasets, highlighting model robustness and stability under varied monitoring environments.

However, most existing datasets still fail to capture sufficient diversity in terms of temporal dynamics, illumination conditions, and contextual variations, particularly within school environments. To bridge this gap, this study introduces RWF-2000+, an extended version of the original dataset designed to enhance data diversity, density, and realism, thereby improving applicability to educational surveillance scenarios.

B. Deep Learning Models for Violence Detection

The advancement of deep learning has enabled diverse approaches for extracting spatiotemporal representations in video-based violence recognition, ranging from hybrid CNN–3D CNN pipelines to lightweight architectures designed for real-time surveillance. Khan et al. [11] proposed a three-stage framework that combines lightweight CNNs with 3D CNNs to jointly model spatial and temporal cues, and their cross-domain evaluation reported stable performance across multiple surveillance datasets. Kaur and Singh [10] introduced a transfer-learning-based ensemble that integrates three pretrained backbones with a stacked LSTM, achieving strong results on both RLVS and RWF-2000. Along a similar direction, Aung et al. [12] presented a dual-stream model that fuses human pose and motion dynamics via weighted fusion, improving recognition performance on RWF-2000.

Several works have further emphasized efficiency and deployability. Zhu and Hu [13] proposed a lightweight two-stream spatiotemporal network deployed in a collaborative cloud–edge architecture to reduce computational cost while maintaining competitive accuracy on RWF-2000. Wang et al. [14] developed an EfficientNet-B0-based model augmented with Bi-LTMA and TSM mechanisms to balance accuracy and runtime efficiency for surveillance devices. Dündar et al. [15] proposed a shallow 3D CNN with only 54.6K parameters, demonstrating that compact architectures can achieve reasonable performance across RWF-2000, RLVS, and Hockey Fight datasets.

Beyond RGB-based spatiotemporal CNNs, recent studies have explored skeleton/graph representations to improve robustness under challenging surveillance conditions. Tran et al. [16], [17] model skeleton sequences as spatiotemporal graphs and leverage GCN-based designs (with attention mechanisms) to capture joint dynamics for violence recognition on benchmarks such as RWF-2000. Complementary to algorithmic advances, system-oriented studies also emphasize practical deployment aspects; for example, Jeevan and Avanthika [4] present an intelligent video surveillance framework integrating real-time detection modules and post-processing to reduce false alarms.

While recent deep learning–based approaches have reported strong performance in violence detection, many rely on computationally intensive or multi-stream architectures that are difficult to deploy in real-time surveillance settings. Even among studies that explicitly consider efficiency or edge-awareness, evaluations are frequently conducted on cloud GPUs or proxy platforms rather than under realistic

embedded constraints. As a result, a gap remains between reported recognition performance and practical deployability, particularly for school surveillance scenarios where low latency, limited resources, and continuous operation are critical.

III. Extended Violence Behavior Dataset

Existing violence detection datasets, such as RWF-2000, have been widely used to evaluate deep learning models; however, their applicability to school surveillance scenarios remains limited. Most videos were collected from public spaces, resulting in differences in context, illumination, and camera viewpoints compared with typical school environments. Moreover, the violent behaviors captured are often large-scale and spontaneous, which do not fully reflect the subtle and short-duration interactions commonly observed in educational settings. These factors can hinder the effective generalization of models in school-oriented surveillance applications.

A. Scientific Basis for Dataset Expansion

Although RWF-2000 is a widely used benchmark for violence detection, its visual context and interaction patterns differ from those typically observed in school surveillance. Most clips are captured in outdoor or semi-public environments (e.g., streets and commercial areas), resulting in camera viewpoints, illumination, crowd density, and interpersonal distances that may not transfer well to indoor school settings. In contrast, school surveillance often involves constrained spaces, more frequent close-range interactions, and subtle physical contacts that can challenge violence classifiers and increase false alarms.

To improve contextual relevance and support more reliable evaluation in educational scenarios, we introduce an additional dataset, termed *RWF-2000+*. RWF-2000+ is designed to (i) broaden viewpoint and illumination diversity representative of indoor school cameras, (ii) include school-like interaction patterns through carefully scripted, safety-supervised enactments, and (iii) enable evaluation under conditions that resemble continuous, real-time school monitoring (e.g., fixed camera placement and typical indoor backgrounds).

The supplementary samples were collected in controlled indoor school environments using camera placements consistent with common classroom surveillance setups. Recordings focus on mild physical interactions performed by student volunteers under supervision to ensure safety and ethical compliance. While these staged scenarios cannot fully capture the unpredictability of real incidents, they provide additional school-oriented motion patterns and background cues that complement the original RWF-2000. Therefore, RWF-2000+ is intended to augment–not replace–RWF-2000 by offering a more school-relevant testbed with increased contextual diversity.

B. Architecture and Organization of the RWF-2000+ Dataset

The expanded RWF-2000+ dataset contains approximately 2,540 short surveillance clips annotated into two classes, *Fight* and *Non-Fight*. Each clip is recorded at 25 frames per second (fps) with an average duration of 4–5 seconds, which provides standardized segments for clip-level benchmarking and captures short-term motion cues relevant to violent interactions in practical surveillance settings. Importantly, this duration does not represent the decision latency of streaming models: MoViNet-Stream produces frame-progressive predictions and can, in principle, trigger an alert before the end of the clip once sufficient evidence is observed. A dedicated early-warning protocol to quantify the minimum observation window for stable alerting (e.g., sub-second triggering time) will be included in an extended version of this benchmark.

RWF-2000+ combines two complementary sources. The first component is the original RWF-2000 benchmark, consisting of 2,000 real-world surveillance videos collected from public areas (e.g., streets, shopping malls, and train stations). These clips typically depict spontaneous and unstructured events and exhibit substantial variability in viewpoint, illumination, and crowd density. The second component consists of 540 supplementary clips newly recorded in controlled, school-like indoor environments with camera viewpoints consistent with classroom surveillance setups. These recordings focus on school-relevant interactions performed by student volunteers under supervision to ensure safety and ethical compliance.

In the supplementary RWF-2000+ recordings, the *Non-Fight* class focuses on close-proximity but non-violent interactions that frequently occur in school environments, including handshakes, conversations at short interpersonal distance, and mild incidental contacts (e.g., brief shoulder/arm bumps). These cases are intentionally included as challenging negatives because their spatial proximity and short-term motion can resemble pre-fight cues and may lead to false positives in real-time surveillance.

Together, the supplementary data increases contextual relevance for educational scenarios while preserving the real-world diversity provided by the original RWF-2000 subset.

Table I
Dataset statistics and split configuration.

Dataset	Train	Val.	Test	Total	Fight	Non-Fight
RWF-2000	1400	300	300	2000	1000	1000
RWF-2000+	1778	380	382	2540	1270	1270

Both datasets are divided into training, validation, and testing subsets as summarized in Table I, maintaining a balanced 1:1 ratio between the *Fight* and *Non-Fight* classes. By combining diverse public-scene videos with school-oriented simulated data, RWF-2000+ extends the contextual diversity of the original benchmark and provides a more representative evaluation setting for real-world school surveillance applications.

C. Data Collection Setup and Environmental Characteristics

The supplementary RWF-2000+ videos were recorded with a handheld smartphone placed at elevated positions

Figure 1. Example frames from the RWF-2000+ dataset captured in controlled school-like environments.

to approximate the viewpoints of classroom surveillance systems. This configuration enables the capture of relevant spatial layouts and motion patterns while maintaining a fully controlled recording environment.

Data collection was conducted under varying illumination conditions within classroom-like settings to introduce visual diversity representative of typical school surveillance scenarios. All recording sessions were closely supervised to ensure safety, consistency, and ethical compliance.

Each simulated scenario involved a small group of participants enacting mild physical interactions such as brief confrontations or pushing commonly observed in educational contexts. Representative examples are shown in Figure 1, illustrating variations in lighting conditions and subject density.

Although the videos were captured using a mobile device rather than fixed CCTV cameras, the camera placements and viewing angles were selected to resemble school surveillance perspectives. This controlled setup provides a reliable testbed for evaluating model robustness under school-oriented visual conditions, serving as an intermediate evaluation setting between controlled simulations and real-world surveillance scenarios.

IV. Systematic Benchmark and Evaluation Method

A. Overview of the Evaluation Framework

This study presents a systematic benchmarking framework to evaluate deep learning models for school violence detection, with a particular focus on the trade-off between recognition accuracy and computational efficiency. Instead of proposing new architectures, the framework aims to provide a fair and consistent comparison across representative models by using a unified training and evaluation protocol, while preserving each architecture's canonical input configuration (e.g., default input resolution and preprocessing).

Two datasets, RWF-2000 and the school-oriented RWF-2000+, are employed to assess both generalization and deployment relevance. An overview of the evaluation pipeline, covering training, testing, and edge deployment, is illustrated in Fig. 2.

B. MoViNet-Stream Architecture and Variant Efficiency Analysis

MoViNet (Mobile Video Network) [18] was selected as a representative lightweight architecture due to its design explicitly targeting efficient video recognition under resource constraints. Compared to conventional 3D CNNs, MoViNet introduces architectural features that are well aligned with real-time school surveillance requirements:

- **Streaming inference**: temporal states are preserved across frames, enabling continuous video processing with low latency.
- **Temporally separable convolutions**: reduce computational cost while maintaining sensitivity to short-term motion dynamics.
- **Edge-aware design**: optimized for deployment on embedded platforms with limited compute resources.

To analyze the accuracy–efficiency trade-off, two representative variants were evaluated. MoViNet-A0-Stream emphasizes inference speed and low resource consumption, whereas MoViNet-A2-Stream provides increased representational capacity at higher computational cost. These variants capture two ends of the efficiency spectrum relevant to real-time surveillance systems.

C. Video Preprocessing and Input Configuration

All models take RGB frame sequences as input. Each input video is represented as a fixed-length clip of consecutive frames to capture spatiotemporal dynamics relevant to violent behaviors. For evaluation, clips are extracted as contiguous frame sequences to preserve motion continuity and ensure consistent clip-level measurement across models.

Input resolutions follow the canonical configuration of each model architecture rather than enforcing a single uniform spatial resolution. Specifically, MoViNet-A0-Stream adopts 172×172, MoViNet-A2-Stream uses 224×224, C3D-based models use 112×112, and X3D-XS + CoAtNet-0 uses 224×224. For all models, the frame rate is fixed at 25 fps to ensure consistent temporal sampling.

During training, clips are randomly sampled along the temporal axis and basic geometric augmentations are applied to improve generalization under varying surveillance conditions. During testing, we disable random temporal sampling and use a deterministic clip extraction strategy to ensure reproducible comparisons.

D. Training Setup and Evaluation Protocol

All MoViNet-Stream models were initialized with pre-trained weights from the Kinetics-600 dataset and fine-tuned independently on the RWF-2000 and RWF-2000+ datasets under a unified training protocol.

The task was formulated as a binary classification problem (Fight vs. Non-Fight). Model performance was evaluated using complementary metrics, where Top-1 accuracy reflects

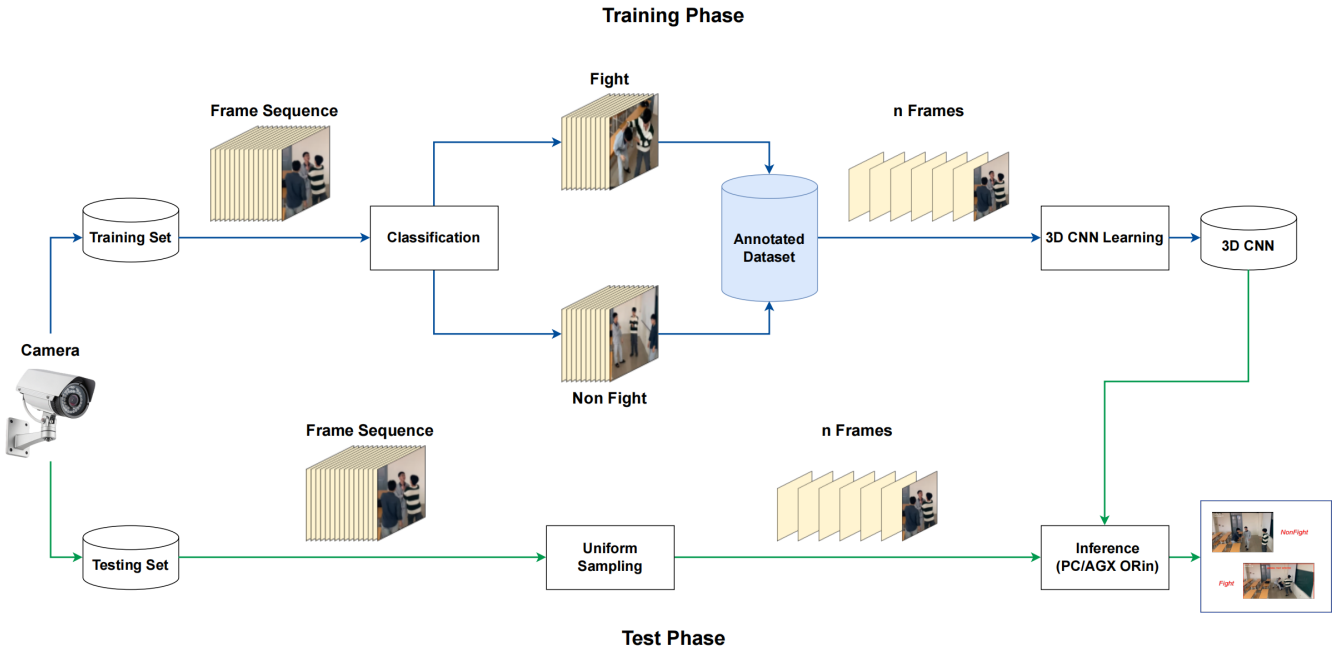

Figure 2. Overview of the proposed experimental evaluation framework for video-based school violence detection.

Table II
PERFORMANCE COMPARISON OF DIFFERENT MODELS ON THE RWF-2000 AND RWF-2000+ DATASETS.

Dataset	Model	Pre-trained	Resolution	Acc	Precision	Recall	GFLOPs	Params
RWF-2000	X3D-XS + CoAtNet-0	Kinetics-400 + ImageNet-1K	224×224	77.59	70.29	95.30	9.23G	31.09M
	C3D + EfficientNet-B0	ImageNet-1K	112×112	70.56	70.19	71.14	38.91G	45.17M
	MoViNet-A0-Stream	Kinetics-600	172×172	89.67	90.50	88.70	1.8G	1.9M
	MoViNet-A2-Stream	Kinetics-600	224×224	**92.00**	**97.01**	86.67	4.9G	4M
	C3D + LSTM	–	112×112	87.75	74.00	95.00	31.60G	16.14M
RWF-2000+	X3D-XS + CoAtNet-0	Kinetics-400 + ImageNet-1K	224×224	84.81	77.36	98.42	9.23G	31.09M
	C3D + EfficientNet-B0	ImageNet-1K	112×112	70.15	69.94	70.68	38.91G	45.17M
	MoViNet-A0-Stream	Kinetics-600	172×172	**92.15**	**95.48**	88.48	**1.8G**	**1.9M**
	MoViNet-A2-Stream	Kinetics-600	224×224	**95.29**	94.82	**95.81**	4.9G	4M
	C3D + LSTM	–	112×112	90.00	87.00	82.01	31.60G	16.14M

Table III
CLIP-LEVEL RUNTIME EFFICIENCY COMPARISON ON RWF-2000 AND RWF-2000+.

Dataset	Model	Inference (ms)	Total Time (ms)	Clips/s
RWF-2000	X3D-XS + CoAtNet-0	13.90	28.00	35.71
	C3D + EffNet-B0	11.06	23.00	43.48
	MoViNet-A0-Stream	**20.77**	**23.59**	**42.39**
	MoViNet-A2-Stream	34.24	38.16	26.20
	C3D + LSTM	33.99	36.60	27.32
RWF-2000+	X3D-XS + CoAtNet-0	13.90	28.00	35.71
	C3D + EffNet-B0	11.06	23.00	43.48
	MoViNet-A0-Stream	**21.64**	**24.30**	**42.91**
	MoViNet-A2-Stream	34.10	38.01	26.31
	C3D + LSTM	33.99	36.60	27.32

overall classification performance, and precision and recall characterize false-alarm and miss tendencies in surveillance settings. Model efficiency was assessed in terms of parameter count, GFLOPs, and clip-level runtime.

For benchmarking, MoViNet-Stream was compared with representative baseline architectures, including X3D-XS + CoAtNet-0, C3D + EfficientNet-B0, and C3D + LSTM. We used a unified training and evaluation *pipeline* across models, and each model was evaluated under its canonical input configuration (as listed in Table II).

V. EXPERIMENTAL EVALUATION

A. Experimental Environment

Model training and accuracy-oriented evaluation were conducted on Google Colab Pro using an NVIDIA Tesla T4 GPU. This cloud environment provides a controlled and reproducible setup for optimization and enables a fair comparison across the evaluated architectures.

All models were trained and evaluated under a unified protocol, including the same dataset splits, training schedule, and reporting metrics. To avoid unintended handicapping,

each architecture was executed under its canonical input setting (i.e., default input resolution and clip configuration), as summarized in Table II. Top-1 accuracy was used as the primary metric for the binary *Fight/Non-Fight* classification task. Model efficiency was analyzed using parameter count and GFLOPs.

Deployment-oriented runtime measurements were conducted separately on an NVIDIA Jetson AGX Orin edge device (Section V-C). While the cloud platform is used for accuracy benchmarking under controlled conditions, Jetson AGX Orin is used to assess clip-level inference latency and throughput under practical embedded constraints, supporting real-time on-device operation under the tested configuration.

B. Performance Evaluation

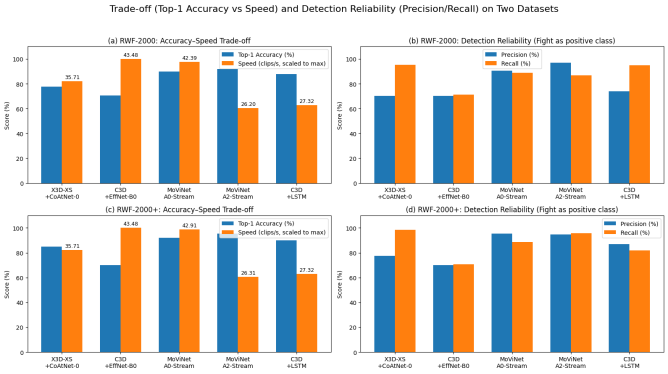

Figure 3. Accuracy–speed trade-off and detection reliability of evaluated models on RWF-2000 and RWF-2000+. (a,c) Top-1 accuracy vs. inference speed (throughput); (b,d) Precision and Recall (Fight as the positive class). For visualization, speed is normalized within each dataset, while the corresponding absolute throughput (clips/s) is annotated; absolute runtime results are reported in Table III.

Table II and Table III, together with Fig. 3, summarize recognition performance and clip-level runtime efficiency across the evaluated models. Overall, lightweight architectures, particularly the MoViNet-Stream variants, provide a strong accuracy–efficiency trade-off compared with conventional 3D CNN and hybrid baselines under the tested configuration.

On the RWF-2000 dataset, MoViNet-A2-Stream achieved the highest accuracy (92.00%) with a computational cost of 4.9 GFLOPs. MoViNet-A0-Stream offers a more deployment-oriented operating point, reaching 89.67% accuracy at only 1.8 GFLOPs while sustaining 42.39 clips/s. In contrast, C3D + LSTM attained 87.75% accuracy but required substantially higher compute (31.60 GFLOPs) and achieved a lower throughput (27.32 clips/s), making it less favorable for real-time deployment.

On the extended RWF-2000+ dataset, performance changes are model-dependent. MoViNet-A2-Stream further improved accuracy to 95.29%, whereas MoViNet-A0-Stream maintained among the highest throughput values (42.91 clips/s), remaining competitive for low-latency deployment. Notably, X3D-XS + CoAtNet-0 also exhibits a noticeable accuracy gain on RWF-2000+, but its computational cost remains considerably higher than that of MoViNet-A0-Stream. Since the input configuration and evaluation pipeline are unchanged, runtime remains largely consistent across datasets, and minor differences are attributed to measurement variability.

Table IV
CLIP-LEVEL INFERENCE PERFORMANCE OF MOVINET-A0-STREAM ON JETSON AGX ORIN. LATENCY IS MEASURED PER INPUT CLIP AND THROUGHPUT IS REPORTED IN CLIPS/S.

Framework	Model	Accuracy (%)	Latency/Clip (ms)	Clips/s
PyTorch (FP32)	MoViNet-A0-Stream	91.99	68.76	14.54
TensorRT (FP16)	MoViNet-A0-Stream	90.05	14.08	71.02

Fig. 3 visualizes the accuracy throughput trade-off across model families, while precision and recall in Table II provide complementary evidence on detection reliability, indicating different trade-offs between false alarms and missed detections.

Overall, these results suggest that streaming-capable architectures with temporally separable convolutions can capture short-duration motion patterns efficiently, supporting low-latency inference for school-surveillance settings under the tested configuration.

C. Edge AI Deployment on NVIDIA Jetson AGX Orin

To characterize on-device inference efficiency under embedded constraints, we benchmarked MoViNet-A0-Stream on an NVIDIA Jetson AGX Orin (32GB LPDDR5, 8-core Arm Cortex-A78AE CPU, and an Ampere GPU with Tensor Cores). Unless otherwise noted, all runtime measurements in this section were obtained directly on the Jetson platform.

MoViNet-A0-Stream was chosen for deployment evaluation because it offers a favorable accuracy–efficiency trade-off in our offline benchmark while incurring substantially lower compute cost than larger variants. Given the higher parameter count and GFLOPs of MoViNet-A2-Stream, its throughput is expected to be lower on embedded hardware; therefore, we report on-device results for A0 as the primary deployment-oriented operating point.

For deployment, the model was exported to ONNX and optimized with TensorRT using FP16 to leverage mixed-precision acceleration on Jetson. As summarized in Table IV, the PyTorch FP32 implementation requires 68.76 ms per clip (14.54 clips/s), whereas TensorRT FP16 reduces latency to 14.08 ms per clip and increases throughput to 71.02 clips/s. The corresponding accuracy changes modestly (91.99% vs. 90.05%) under the same evaluation setting. Overall, these results show that TensorRT mixed-precision optimization can markedly improve clip-level throughput for the evaluated configuration on Jetson.

Finally, MoViNet's streaming formulation maintains temporal state across consecutive inputs, which can reduce redundant computation relative to non-streaming 3D CNN inference. While this deployment study focuses on clip-level compute performance rather than full system latency (e.g.,

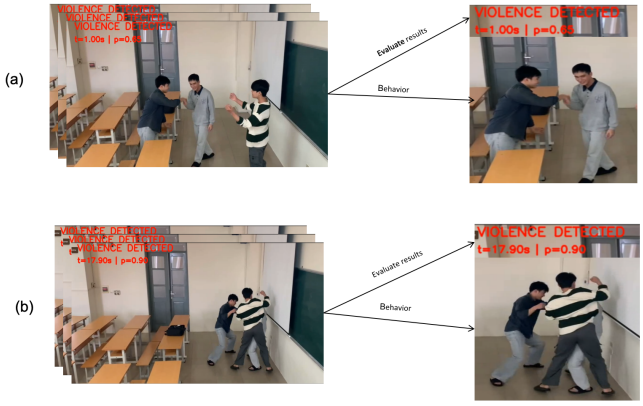

Figure 4. Visual interpretation of prediction outcomes from the MoViNet-Stream model. (a) *False positive*: a handshake or close interaction is misclassified as violence due to minimal interpersonal distance and rapid hand motion. (b) *True positive*: an actual violent incident characterized by strong motion intensity and consistent temporal dynamics.

capture, preprocessing, and decision logic), the measured throughput indicates that on-device inference is unlikely to be a computational bottleneck under our tested settings.

D. Case Study and Error Analysis

To further analyze the contribution of key design components, a qualitative ablation-style analysis was conducted to examine how different modeling factors affect the behavior of the MoViNet-Stream architecture in realistic school surveillance scenarios. Representative prediction cases are shown in Figure 4. First, the impact of streaming-based temporal modeling is observed by comparing predictions under short, transient motions versus sustained motion patterns. In false positive cases (Figure 4a), brief non-violent interactions such as handshakes or close physical contact are occasionally misclassified as violent. This behavior indicates that emphasizing short-term temporal dynamics alone, without sufficient contextual persistence, can cause the model to overreact to rapid but benign movements.

Second, the effectiveness of temporal accumulation across consecutive frames is evident in true positive examples (Figure 4b). When violent actions exhibit consistent and high-intensity motion over time, the streaming mechanism enables the model to integrate temporal evidence progressively, leading to stable and confident predictions. This observation highlights the importance of continuous temporal state propagation in distinguishing genuine violence from momentary interactions.

Third, the influence of training data diversity is reflected in the reduction of false negatives when using the extended RWF-2000+ dataset. Mild or short-lived confrontations that were previously misclassified become easier to detect after exposure to school-specific interaction patterns, suggesting that contextual diversity plays a complementary role to architectural efficiency.

VI. Conclusion and Future Work

This study presented a deployment-aware evaluation of deep learning models for school violence detection, emphasizing the trade-off between recognition accuracy and computational efficiency under real-time constraints. Experiments on RWF-2000 and the school-oriented RWF-2000+ dataset show that lightweight streaming architectures, particularly MoViNet-Stream variants, achieve a favorable balance between accuracy and practical deployability compared with conventional 3D CNN-based baselines. Edge-device measurements on an NVIDIA Jetson AGX Orin further indicate that streaming inference, combined with mixed-precision optimization, enables real-time on-device detection with low computational overhead. Overall, our findings suggest that accuracy-centric evaluation alone is insufficient for surveillance applications, and that deployment-aware benchmarking is essential for real-world school monitoring systems.

Future work will extend RWF-2000+ along two directions. First, we will incorporate a dedicated early-warning evaluation protocol to quantify the minimum temporal observation window required for stable alerting (e.g., sub-second triggering behavior) in streaming settings. Second, we will expand the *Non-Fight* class with additional hard-negative activities (e.g., high-motion actions and prolonged close-contact interactions) to further reduce false alarms under realistic school dynamics. Finally, we will explore multimodal cues (e.g., pose or audio when available) to improve robustness while maintaining real-time performance on embedded devices.

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